MIL-T-36022B
28 June 1978
SUPERSEDING
MIL-T-36022A
13 March 1974 and
MIL-T-36022
5 June 1959

MILITARY SPECIFICATION

TABLE, OPHTHALMIC LABORATORY

This specification is approved for use by all Departments and Agencies of the Department of Defense

1 SCOPE

1.1 Scope This specification covers one type of ophthalmic laboratory table, suitable for field use.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents The following documents of the issue in effect or date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

MM-L-736	- Lumber and Timber Hardwood
NN-P-530	- Plywood, Flat Panel
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, Tube, and Wire, Extruded, 6061
QQ-A-200/9	- Aluminum Alloy Bar, Rod Shapes, Tube, and Wire, Extruded, 6063
QQ-A-225/6	- Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 2024
QQ-A-225/7	- Aluminum Alloy Bar, Rod, and Wire, Rolled, Drawn, or Cold Finished, 5052
QQ-A-250/2	- Aluminum Alloy 3003, Plate and Sheet
QQ-A-250/4	- Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6540

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SPECIFICATIONS

FEDERAL

QQ-A-601 - Aluminum Alloy Sand Castings.
QQ-S-763 - Steel Bars, Snapes, and Forgings - Corrosion Resisting
QQ-S-766 - Steel Plates Sheets and Strip-Corrosion Resisting
QQ-W-423 - Wire, Steel, Corrosion-Resisting
TT-E-527 - Enamel, Alkyd, Lustreless
TT-P-320 - Pigment, Aluminum, Powder and Paste, for Paint
TT-P-636 - Primer Coating, Alkyd, Wood and Ferrous Metal
TT-P-1757 - Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
TT-V-119 - Varnish, Spar, Phenolic-Resin
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-601 - Boxes, Wood Cleated-Plywood
PPP-B-621 - Boxes, Wood Nailed and Lock-Corner
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-1055 - Barrier Material, Waterproofed, Flexible

STANDARDS

AIR FORCE-NAVY AERONAUTICAL

AN4 - Bolt-Machine, Aircraft
AN25 - Bolt-Clevis
AN503 - Screw-Machine Fillister Head Coarse Thread
AN505 - Screw-Machine, Flat Head, 82 Degree, Coarse Thread
AN525 - Screw-Washer Head

FEDERAL

FED TEST METHOD STD NO 151 - Metals Test Methods
FED TEST METHOD STD NO. 595 - Color.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage
MS16686 - Medical Services Symbol.
MS20470 - Rivet, Solid-Universal Head, Aluminum and Aluminum Alloy.

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DRAWING

DEFENSE PERSONNEL SUPPORT CENTER
10044 Table, Ophthalmic Laboratory

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 886 - Outlet Boxes and Fittings for Use in Hazardous Locations.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611, 333 Pfingsten Road, Northbrook, IL 60062, 1285 Walt Whitman Road, Melville, LI, NY 11746, 1655 Scott Blvd., Santa Clara, CA 95050.)

3 REQUIREMENTS

3.1 Materials. Except as specified herein, materials shall conform to the applicable portions of the specifications, standards, and drawing (with bill of materials) shown in Section 2 of this specification, and shall be capable of meeting the requirements shown therein. In addition, the metals shall conform to the chemical requirements of their applicable specifications when tested as specified in 4.4.1. When materials to be used are not specifically designated, those used shall be of the highest commercial quality, and suitable for the purpose intended.

3.1.1 Plywood. Plywood shall conform to the requirements of NN-P-530 and shall be group A or B plywood. Both sides of the plywood shall be finished with a high resin content, high density overlay. The finish shall be olive drab, color 34087, of FED. TEST METHOD STD. NO. 595. The adhesive used in fabricating the plywood shall be fungus and mildew resistant.

3.1.2 Primer, metal. The metal primer shall conform to color Y, composition "G" or "L", as applicable, of TT-P-1757.

3.1.3 Primer, wood. The wood primer shall conform to TT-P-636.

3.1.4 Enamel. The enamel shall conform to TT-E-527, color 34087 of FED. TEST METHOD STD. NO. 595, except that for the Medical Services Symbol, color 37038 of the same standard shall be used.

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3.2 Component parts The ophthalmic laboratory table shall be a combination table and stool, approximately 48 inches long, 26 inches wide, 33-9/16 inches high, from floor to tabletop, and 42-3/16 inches overall height. The stool shall be an integral part of the table, and so designed and located on the table, that it may be swung under the table when not required The table shall consist of an upper section, a top shelf, a removable lower section, a stool assembly and a leg assembly fabricated of aluminum alloy, except for the tabletop, the cover of the lower section, and the seat of the stool

3.2.1 Upper section The upper section shall consist of a plastic-coated plywood tabletop, tabletop frame, four leg brackets, two tray guide supports, and two tray guides. The tabletop shall be provided with a work tray The work tray shall consist of the work tray body, two side tray rims and two long tray rims, two tray guides, a tray stiffener, and two pull stops. There shall also be two leg locking arms and springs

3.2.2 Top shelf The top shelf shall consist of a top shelf and a top shelf back brace.

3.2.3 Lower sections The lower section shall be essentially an assembly of two drawer compartments, two drawers, with a cover forming a shelf, and a stool attached to one of the drawer compartments The section shall consist of two drawer frames, two drawer stiles, eight tie plates, two transverse supports for the lower section, two longitudinal supports for the lower section, two drawer frame covers, a masonite cover for the lower section and an arm locking spring

3.2.3.1 Drawer Each drawer shall consist of a drawer front, a drawer hole cover, a drawer body, a drawer back, a pull stop, six drawer guide supports, two drawer guides and two drawer slides

3.2.4 Stool assembly The stool assembly shall consist of two backing plates for the arm support, an arm support, two washers, an arm support shaft, an arm, a stool shaft, two stool flange shafts, a stool shaft spring, a wood seat, and three screw plugs

3.2.5 Leg assembly The leg assembly shall consist of two long legs two short legs, two leg braces four foot pads, four plugs four stops, four locking screw supports, four locking screws, and four knurled knobs

3.2.6 Miscellaneous components The table shall be provided with a UL approved, 6 outlet terminal strip, motor base plug, and suitable fasteners which conform to UL 886

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3.3 Construction. Construction shall conform in all details to the requirements specified herein and as shown on DPSC Drawing 10044. Construction shall be sound and durable in all respects. Assembly shall be welded on jigs capable of maintaining required tolerances In lieu of the construction shown on the drawing for the assembly of part numbers 4, 27 and 38, and the construction of part numbers 27 and 38 (pages 1, 3, 8 and 11 of the drawing), the following alternate is permitted

Part number 4 shall be eliminated Part numbers 27 and 38, at their juncture, shall be cut at a 45° angle at one meeting edge and at 90° angles at the other meeting edge Part numbers 27 and 38 shall then be butt-welded The outside of the welds shall be ground smooth and even with the adjacent surfaces The inside of the welds shall be finished smooth and free of sharp edges and burrs.

3 3 1 Welding Welding shall be done by qualified welders using appropriate aluminum welding equipment in a qualified process. Spot welds shall be capable of developing the tensile strength of the parent metal. All exposed welds shall be sufficiently smoothed to blend with the adjacent surfaces after the enamel finish specified in 3.4 has been applied All other welds shall be finished in a clean and proper manner All welds shall be clean, nonporous, and sound

3 3 2 Grounding tag A white fabric tag of suitable size shall be attached to the wiring system The tag shall be legibly and permanently marked with the legend "Before use, the table should be grounded by any suitable means "

3 4 Finish The table shall be finished after welding has been completed but before attachment of tabletop, top shelf, etc (final assembly), as specified herein

3 4 1 Aluminum alloy All aluminum alloy surfaces shall be free of dirt, oil, grease, welding flux and loose scale Surfaces shall be thoroughly cleaned with solvents, detergents, or other processes, which have no deleterious effect on the surfaces After a thorough cleaning and degreasing, surfaces to be enameled (all accessible aluminum surfaces except the working surfaces of the drawer slides and guides) shall be given a priming treatment using primer specified in 3 2.2, thinned, applied by spray and air-dried for not less than 30 minutes before topcoating Topcoating shall be accomplished with one coat of enamel specified in 3 1.4, applied by spray and cured by a heat treatment equivalent to baking at 250°F (121°C) for not less than 45 minutes

3 4.2 Tabletop All exposed edges of the tabletop shall be sealed before attachment of tabletop to table as follows

Any splits showing on edges shall be plugged with a good commercial type crack filler, which shall be touch-sanded smooth before the edges of the panels are sealed. The edges of the panels shall be sealed with two coats of a spar varnish highly pigmented with flake aluminum, to resist moisture

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absorption. This moisture barrier shall be formulated in the following proportions: One gallon of phenolic varnish conforming to TT-V-119, pigmented with 1-1/2 pounds of aluminum paste conforming to type II, class B of TT-P-320. At least 24 hours drying time shall be accorded each coat. After sealing, the edges shall be given a coat of primer (see 3 1 3). As an alternate, a polyvinyl plastic edge molding shall be used in lieu of the painted edge. The molding shall be permanently and tightly secured to the tabletop edge.

3 4.3 Assembly After final assembly all surfaces of the table, except the masonite top of the lower section, which shall be unfinished, and the seat of the stool (see 3.4.4), shall be given a second coat of enamel as specified in 3 1 4 The electrical receptacles shall be adequately protected during application of the second coat of enamel so that their intended function is not impaired by paint, and after finishing shall pass the continuit test (see 4 4 2). The working surfaces of the drawer slides and guides shall b lubricated for ease of operation The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, crocking, acid, alkali, and solvent As an alternate, the tabletop, conforming to 3.2.1, providing a smooth uniform surface resistant to chipping, abrasion and hydrocarbons, shall not be painted

3 4.4 Finish of seat The seat shall be sanded smooth, filled, and stained a dark oak color, followed by a coat of sealer The seat shall then be sanded smooth, a coat of spar varnish applied, resanded and given a final varnish application

3 4.5 Film thickness of paint Contractor shall endeavor to control film thickness so that the thickness of a single dried coat of paint is 0.3 to 0 7 mil for primer and 1 0 to 1 5 mil for enamel

3.5 Identification marking Each table shall have a nameplate which shall be as specified in DPSC Drawing 10044 and bill of materials, and shall bear the information shown thereon The nameplate shall be attached to the table, in the manner and location shown on the drawing, after painting has been accomplis In addition to the nameplate, each table shall be marked with a Medical Service Symbol in the size and location shown or the drawing. Enamel specified in 3 1 4 shall be used for marking

3 6 Workmanship Workmanship shall be first class throughout Drawers, tray, and table itself, shall be square and free from twist or distortion All moving parts shall operate smoothly and easily Tables shall be free of sharp edges, burrs, splinters, or any other defects which detract from their appearance or impair their serviceability

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality supplied by the manufacturer of the component or material may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table I. The unit of product for examination purposes shall be one table.

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TABLE I. Sampling for examination.

Type of defect	Inspection level	AQL (percent defective)
Major	S-2	1 5
Minor	S-2	2 5
Dimensional	S-1	4 0

4 2 2 For tests. Sampling for tests shall be in accordance with MIL-STD-1 and as indicated in table II

TABLE II Sampling for tests.

Characteristic	Sample unit	Inspection level	AQL (percent defective)
Base metal	4 ounces from each lot of each metal used	--	1/
Continuity test	one table	II	1/

1/ Acceptance number shall be zero

4 2 3 Components Sufficient samples of the components shall be selected from each lot of material used in the manufacture of the tables to perform tests as required by the applicable subsidiary specifications or by tests specified herein When sample size for components covered by the subsidiary specifications is not specified, the sample size for each test shall be as indicated above No test result shall fall below the minimum or exceed the maximum values specified The sample unit for testing shall be the quantity necessary to perform all required tests one time each Where possible, the same sample unit shall be used for the determination of two or more test characteristics

4 3 Examination The table shall be examined for defects including, but not limited to, those listed in table III

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TABLE III. Classification of defects.

Categories	Defects
Major	
101	Any component missing or not as specified.
102	Table does not rest evenly on flat surface
103	Materials nonconforming to specification
104	Table not free of burrs, sharp edges or splinters on exterior edges.
105	Table does not fold properly.
Minor	
201	Welds not ground smooth.
202	Welds exhibit cracks or porosity
203	Identification marking missing or incorrect
204	Packing or packaging improper.
205	Improper marking.

4.3.1 Dimensions. Any dimension not within the specified tolerance shall be classified as a defect and shall be cause for rejection of the table

4.3.2 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129

4.4 Tests. Tests of materials used in the fabrication of the tables shall be conducted as specified herein. Other tests may be made, as necessary, to determine that the finished tables comply with this specification, and that the materials comply with the applicable portions of the specifications, standards, and drawing listed in Section 2.

4.4.1 Chemical analysis of the metals. The samples selected in accordance with 4.2.2 shall be analyzed in conformance with method 111 or 112 of FED TEST METHOD STD NO 151. In case of dispute, method 111 shall be used for referee analysis

4.4.2 Continuity test. The unit shall be plugged into a nominal 115 volt AC power source. Each outlet shall be tested by means of a test lamp fitted with a standard 2 blade male plug. Upon insertion of the test lamp into each outlet, the lamp shall remain lighted, without failure, for at least 2 minutes

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6 2).

5.1.1 Unit of issue. One table, as specified, constitutes one unit of issue.

5 1.2 Level A

5.1.2 1 Unit preparation Each unit shall be folded for shipping as shown on DPSC Drawing No 10044, page 2 of 13. The stool shall be protected by wrapping in flexible corrugated fiberboard and shall be secured to prevent shifting during shipment and handling All exterior, finished surfaces of the table shall be wrapped with kraft paper and cushioned with flexible corrugated fiberboard or other suitable material Wrapping and cushioning shall be secured with tape Cushioning and wrapping shall be accomplished in such a manner so as to insure against marring, scratching, or damage to the unit.

5 1 3 Level C Each table shall be clean, dry and packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity

5.2 Packing Packing shall be level A, B or C as specified (see 6.2)

5.2.1 Level A Each unit shall be overboxed in an exterior container designed for a type 3 load and constructed in accordance with PPP-B-601, overseas type or PPP-B-621, class 2. The unit shall be blocked, braced and cushioned within the box in such a manner so as to prevent shifting, damage, or marring of surfaces All box panels, except the base, shall be lined with a case liner conforming to PPP-B-1055, class H-2 or L-2(b) As an alternate, the unit shall be protected with a shroud fabricated of material conforming to PPP-B-1055, class H-5 or M-1 The box design shall include skids of not less than 1-1/2 inches in thickness Closure and strapping shall be as specified in the appendix of the applicable box specification

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution

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5.2.2 Level B. Each unit shall be packed in a box designed for a type 2 load, and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Each packaged table shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity These packs shall conform to the applicable carrier's rules and regulations.

5 3 Marking Exterior container shall be marked in accordance with MIL-STD-129

6. NOTES

6 1 Intended use The ophthalmic table is intended for use as a work bench on which to fabricate spectacles for use by military personnel The table is designed for use in the field and to accommodate a laboratory technician while sitting at the bench or standing

6 2 Ordering data Procurement documents should specify the following.

(a) Title, number and date of this specification
(b) Selection of applicable levels of packaging and packing (See 5 1 and 5 2)
(C) National Stock Number

6.3 Availability of color chips Individual 3x5 inch chips of color 34087 are available from:

Commanding General
Aberdeen Proving Ground
Aberdeen, Maryland 21005
ATTN Coating and Chemical Laboratory

6 4 This specification covers the following item appearing in the Federal Supply Catalog

National Stock Number	NOMENCLATURE
6540-00-299-8742	TABLE, OPHTHALMIC LABORATORY

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6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensivenes of the changes.

Custodians
Army - MD
Navy - MS
Air Force - 03

Review activity
DOD - MB

Procuring activity.
DLA-DM

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MIL-T-36022E TABLE OPHTHALMIC LABORATORY

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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